ToyModelSOEMC Training Run: Explanation of All MATLAB Outputs

Training__2x2x2__share package

1 Purpose

This document explains every output produced by the training run script `ToyModelSOEMC_run_training_2x2x2` which launches `ToyModelSOEMC_run.m` with:

- `MINIMAL_MODE = false`
- `TRAINING_MODE = true`

In this configuration, the joint policy search in Step 4 uses a reduced $2 \times 2 \times 2$ grid:

- $d \in \{90, 365\}$ days
- $\phi_R \in \{0, 1.0\}$
- $\phi_{IQ} \in \{0.04, 0.08\}$

2 Explicit Run Outputs Used in This Report

This report is based on the concrete exported artifacts from the run folder:

- `run_outputs/20260719_191707/`
- Manifest timestamp: `2026-07-19 19:20:14`
- Run success flag in manifest: `1`

The six exported figure sets detected in that folder are:

1. `01_Supply_Crisis_Demo.(png|pdf|fig)`
2. `02_Vietnam_Reserve_Adequacy_Under_Growth.(png|pdf|fig)`
3. `03_Energy_Master_Plan_VIII_Coal_Transition.(png|pdf|fig)`
4. `04_Joint_Optimal_Policy_Heatmap_at_d.(png|pdf|fig)`
5. `05_Joint_Optimal_Welfare_vs_Reserve_Level.(png|pdf|fig)`
6. `06_Joint_Optimal_Stock-Drawdown_Interaction.(png|pdf|fig)`

In addition, the policy summary values below are taken from `joint_optimal_results.txt` generated by the same package run.

3 Observed Numerical Outputs (from joint_optimal_results.txt)

Metric	Value
d^* (days)	90
ϕ^*_R	0.50
ϕ^*_{IQ}	0.04
R^*_{oil}	0.0740
R^*_{coal}	0.1110
$E[W]_{opt}$	-44.1433
$E[W]_{zero}$	-44.1433
Storage cost (% of C)	4.444
$\lambda_{benefit}$ (%)	0.000
C_{ss}	0.1313
Y_{ss}	0.1626
Q^*_{oil}	0.0650
Q^*_{coal}	0.0650
N_{MC}	40
N_d	2

4 Output Inventory (At a Glance)

The MATLAB run produces four categories of outputs:

1. Console output (progress, diagnostics, policy summaries)
2. Interactive figures (scenario and policy plots)
3. Saved data/text artifacts (result files)
4. Workspace state updates (updated model parameters and arrays)

5 Console Outputs and Meaning

Step 1: Dynare compilation and initialization

Printed header: `=== STEP 1: Running Dynare ===`
What it means:

- Dynare parses `ToyModelSOEMC.mod`.
- Model objects `M_`, `options_`, `oo_` are created.
- The custom steady-state file `ToyModelSOEMC_steadystate.m` is used.

Step 2: Steady-state check

Printed header: `=== STEP 2: Steady-State Check (Vietnam calibration) ===`
Key printed values:

- Y_{ss}, C_{ss}, r_{ss}, E_{ss}, NX_{ss}
- Ratios such as C_{ss}/Y_{ss} and I_{ss}/Y_{ss}
- Commodity-level reserve and capacity diagnostics

How to interpret:

- Confirms calibration consistency before policy simulations.
- Checks whether steady-state moments are plausible for the intended Vietnam setting.

Step 3: Deterministic crisis demo

Printed header: `=== STEP 3: Deterministic Crisis Demo ===`

Key printed lines:

- Crisis specification (supply disruption path)
- Scenario welfare values W for each ϕ_R scenario

How to interpret:

- Demonstrates reserve drawdown policy effect under a known shock path.
- The relative W values summarize which drawdown rule performs better in that deterministic setup.

Step 3b: Growth adequacy scenario

Printed header: `=== STEP 3b: Vietnam Growth Scenario ===`

Key printed lines:

- Current reserve coverage (days)
- IEA benchmark (90 days)
- End-horizon coverage under fixed reserves

How to interpret:

- Shows that with GDP/import growth, fixed nominal reserves imply declining coverage ratios.
- Supports policy logic of scaling reserve targets with demand.

Step 3c: Energy Master Plan transition scenarios

Printed header: `=== STEP 3c: Energy Master Plan VIII -- Coal Transition Scenario ===`

Key printed lines:

- Scenario labels and welfare values W
- Qualitative interpretation block (A/B/C scenario comparison)

How to interpret:

- Compares no-policy, investment, and reserve-buffer responses during coal phase-down.
- Highlights interaction between investment and reserves as transition tools.

Step 4: Joint optimization

Printed header: `=== STEP 4: Joint Optimal Policy (phi_R x phi_IQ x d) ===`

Key printed lines:

- Active policy grids and Monte Carlo count
- Per-grid-point expected welfare values
- Final optimal tuple $(d^*, \phi_R^*, \phi_{IQ}^*)$

How to interpret:

- This is the central optimization result for policy recommendation.
- In training mode, result quality is illustrative because grid and run count are reduced.

6 Figures Produced by MATLAB and Interpretation

Figure name (window title)	**What it shows and how to read it**
`Supply Crisis Demo`	Time paths for consumption, energy services, and reserve stocks under deterministic disruption. Compare scenario lines to see buffering strength of policy settings.
`Vietnam: Reserve Adequacy Under Growth`	Coverage (days) over time as demand grows. The gap between fixed-reserve line and target lines visualizes adequacy erosion if stock does not scale.
`Energy Master Plan VIII -- Coal Transition`	Consumption, energy-service, and coal-stock trajectories for transition scenarios A/B/C. Quantifies trade-offs among no response, higher investment, and reserve use.
`Joint Optimal: Policy Heatmap at d*`	Heatmap of expected welfare at optimal reserve level across (ϕ_R, ϕ_{IQ}). Brightest region marks best policy combination for that d.
`Joint Optimal: Welfare vs Reserve Level`	Expected welfare as a function of reserve days at optimal (ϕ_R^*, ϕ_{IQ}^*). The peak indicates preferred reserve sizing.
`Joint Optimal: Stock-Drawdown Interaction`	Welfare-versus-d curves for different ϕ_R values, showing complementarity between stock size and drawdown aggressiveness.

7 Embedded Output Figures (Explicitly Included)

This section embeds the concrete figure exports from the run output folder `20260719_191707` directly into the presentation document.

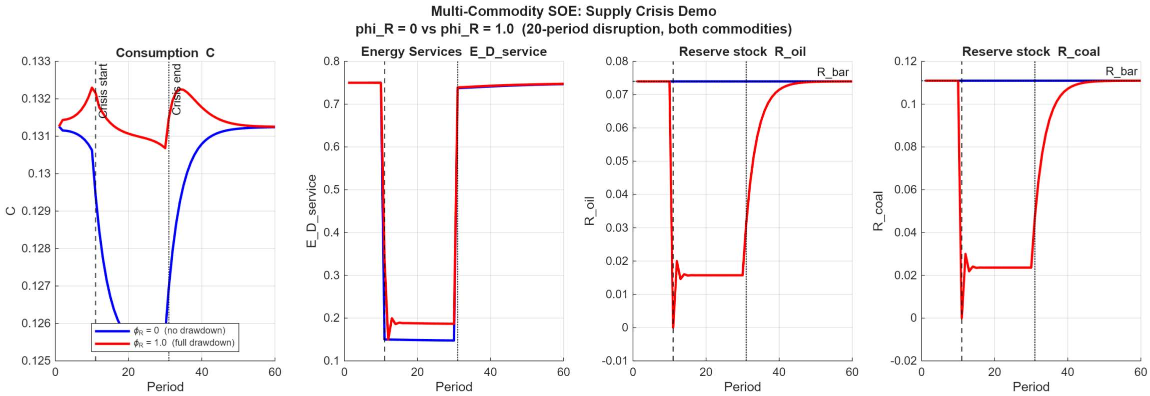

Figure 1: Deterministic crisis comparison of policy responses.

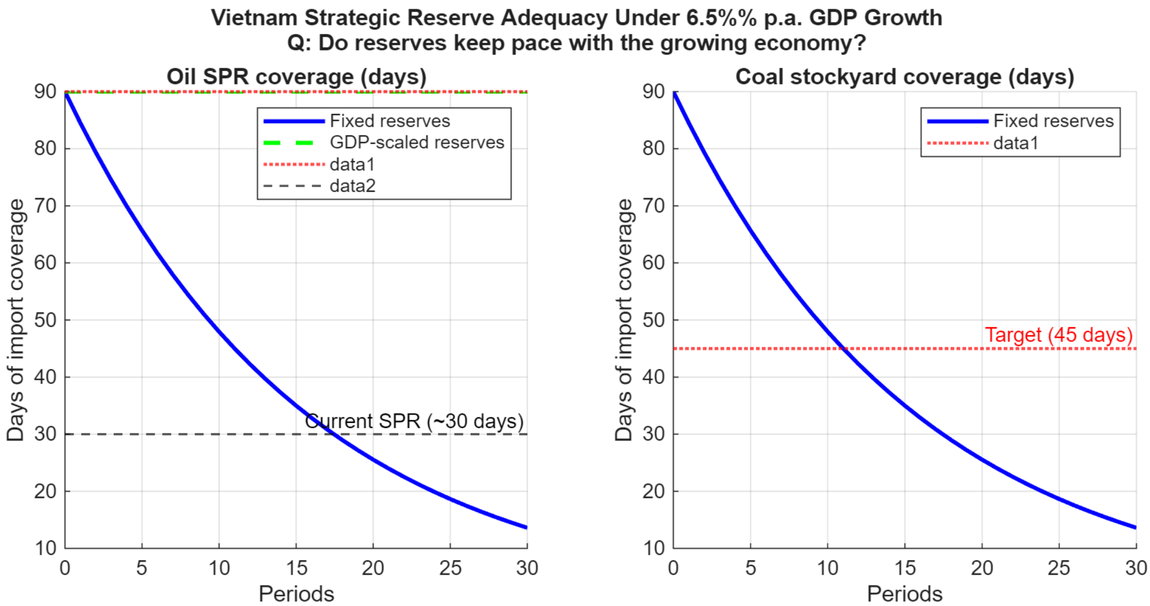

Figure 2: Coverage erosion under GDP/import growth with fixed reserves.

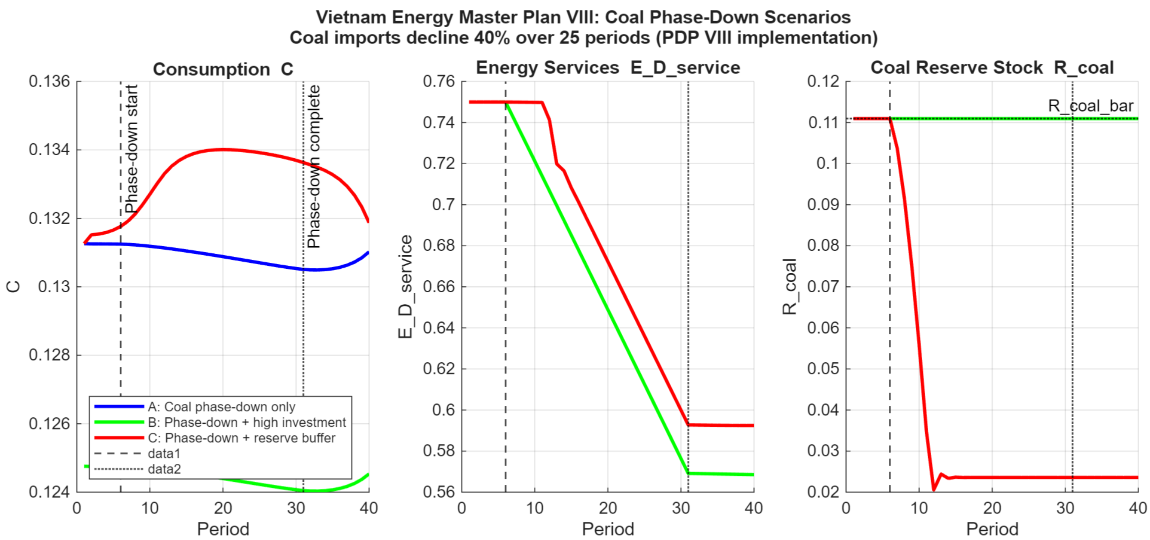

Figure 3: Coal phase-down scenarios and transition dynamics.

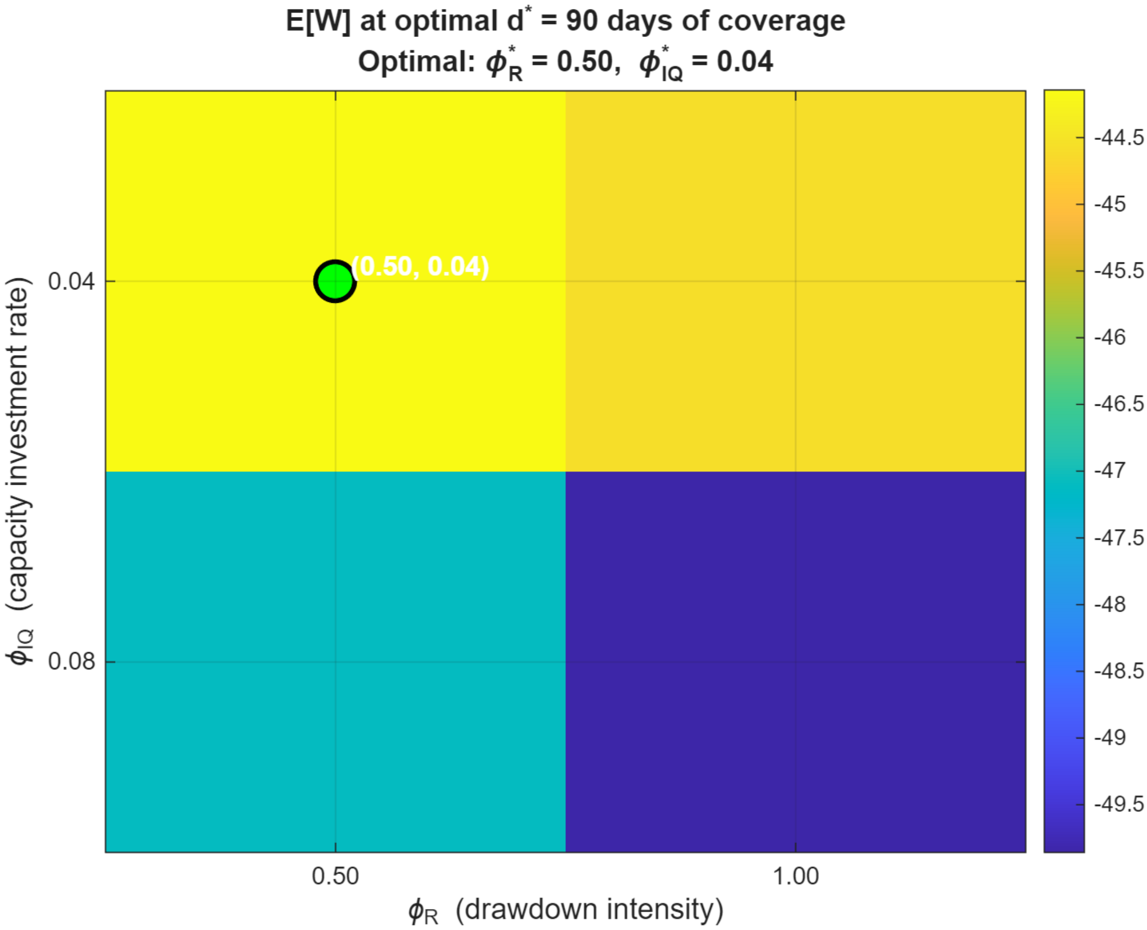

Figure 4: Policy interaction heatmap at the selected reserve level.

Figure 01: Supply Crisis Demo

Figure 02: Reserve Adequacy Under Growth

Figure 03: Energy Master Plan VIII Transition

Figure 04: Joint Optimal Heatmap

Figure 05: Welfare vs Reserve Level

Figure 06: Stock–Drawdown Interaction

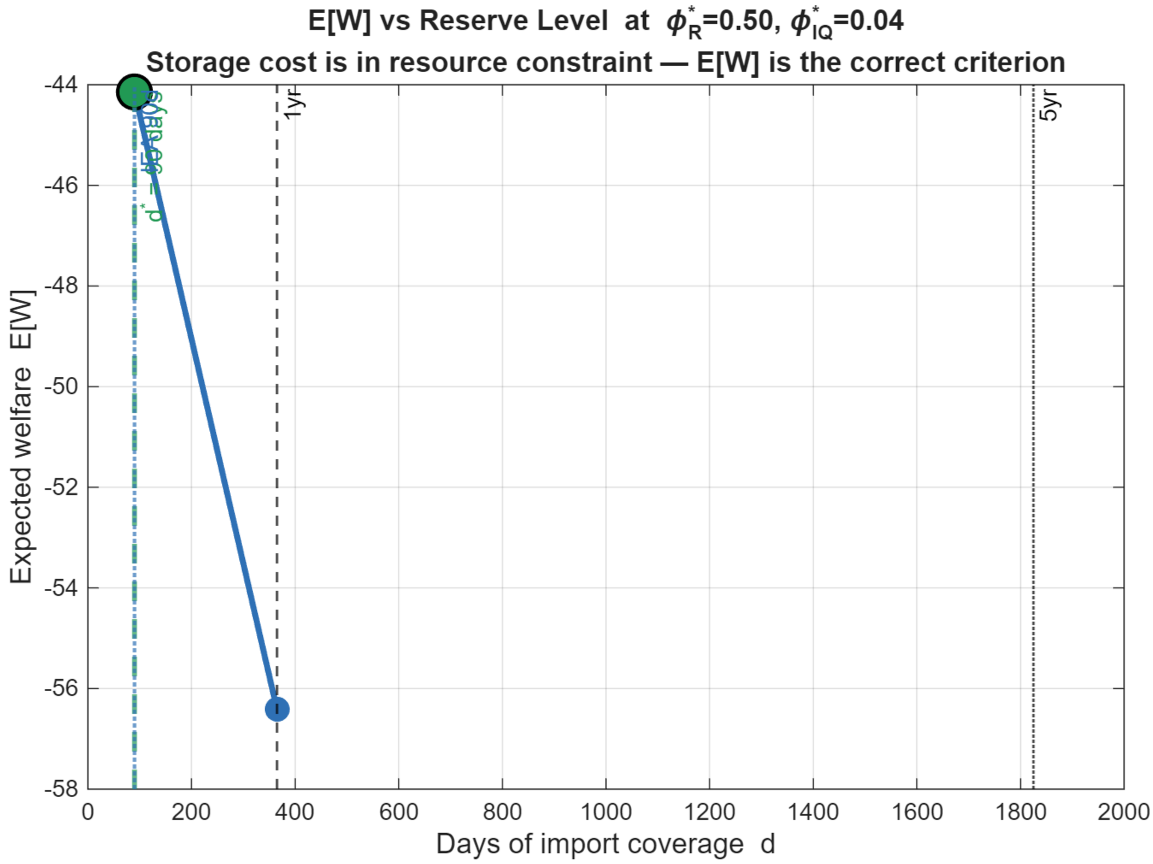

Figure 5: Expected welfare profile over reserve coverage days.

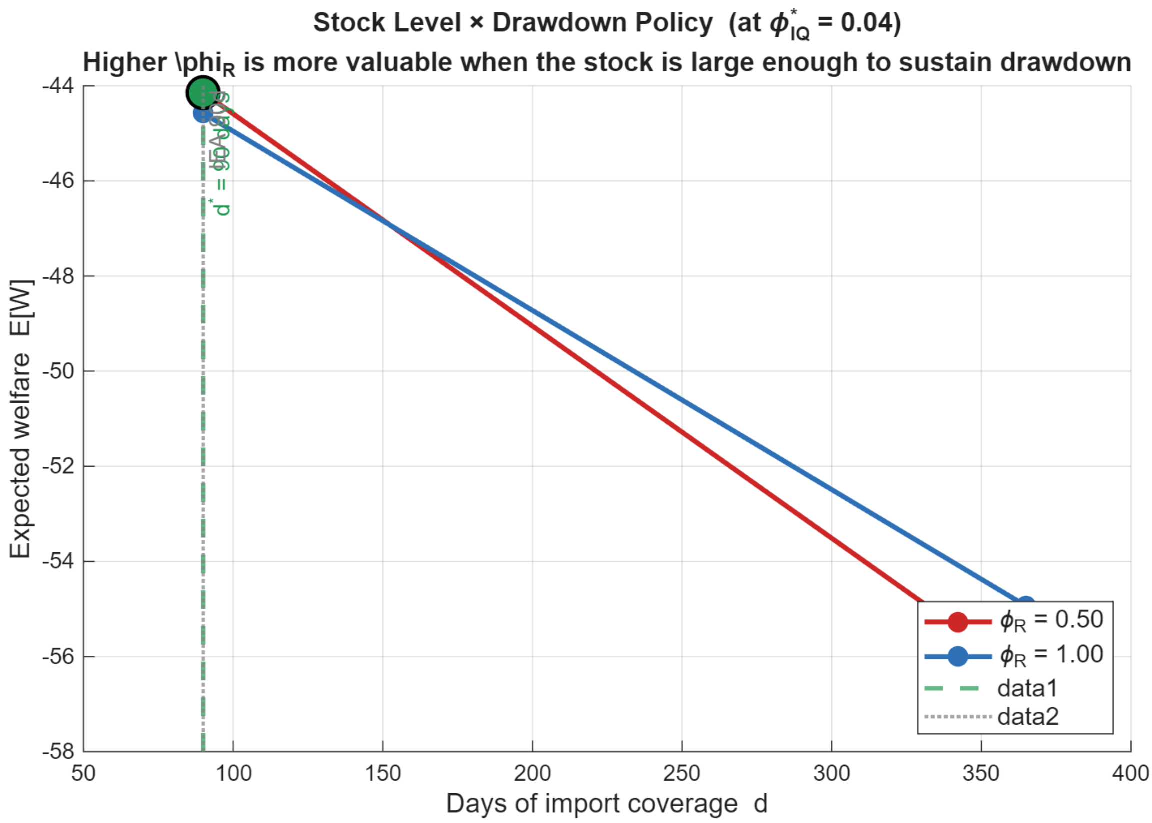

Figure 6: Complementarity between reserve stock size and drawdown aggressiveness.

8 Saved Files and Their Content

joint_optimal_results.txt

Produced by ToyModelSOEMC_joint_optimal.m. Contains:

- Optimal policy tuple: d_star, phi_R_star, phi_IQ_star
- Implied reserve stocks and days coverage
- Welfare metrics: EW_opt, EW_zero
- Storage-cost and welfare-benefit summary fields
- Calibration and run metadata (e.g., N_MC, N_d)

For this explicit run, the file reports:

- d_star=90, phi_R_star=0.50, phi_IQ_star=0.04
- N_MC=40, N_d=2

ToyModelSOEMC_steadystate.mat

Produced by ToyModelSOEMC_steadystate.m. Stores:

- ys: steady-state vector in model variable order
- params: updated parameter vector (including capacity-bar updates)

Complete output-file glossary (clear explanations)

File	Purpose
joint_optimal_results.txt	Plain-text summary of optimal policy and welfare metrics; primary numeric output for reporting.
ToyModelSOEMC_steadystate.mat	MATLAB binary with solved steady state and updated parameter vector.
ToyModelSOEMC.log	Dynare execution log for debugging preprocessing/solve issues.
run_outputs/<timestamp>/matlab_console.log	Captured console output from export run (diary), if available.
run_outputs/<timestamp>/export_manifest.txt	Index of all exported figure files and their paths, with timestamp and success flag.
run_outputs/<timestamp>/NN_<name>.png	Raster figure export for slides/documents.
run_outputs/<timestamp>/NN_<name>.pdf	Vector figure export for high-quality print/LaTeX use.
run_outputs/<timestamp>/NN_<name>.fig	Native editable MATLAB figure format.
run_outputs/<timestamp>/workspace_snapshot.mat	End-of-run workspace snapshot for reproducibility and post-run analysis.
ToyModelSOEMC_outputs_explained.tex	Source of this documentation report.
ToyModelSOEMC_outputs_explained.pdf	Compiled documentation with embedded output figures.

ToyModelSOEMC_outputs_explained.aux/.log/.out/.fls/.fdb_latexmk/.synctex.gz	LaTeX intermediate/build files; safe to regenerate.

9 Workspace Outputs (Not Auto-Saved)

After run completion, MATLAB workspace includes key variables:

- From run script: scenario paths/structures (e.g., results_demo, results_emp)
- From joint optimization: W_joint, phi_R_star, phi_IQ_star, d_star
- Dynare objects: M_, oo_, options_

These are useful for additional analysis but are volatile unless explicitly saved.

10 Why Results Can Differ Across Runs

Some outputs are stochastic because Step 4 uses Monte Carlo shocks in PF mode. Differences arise from:

- Random shock draws in ToyModelSOEMC_shock_gen.m
- Solver convergence path (mitigated with homotopy steps)
- Selected run mode (full vs training vs minimal)

11 Recommended Reporting Practice

For reproducible reporting, save these artifacts after each run:

1. Console log (MATLAB diary)
2. Figure exports (PNG/PDF)
3. joint_optimal_results.txt
4. ToyModelSOEMC_steadystate.mat

This document can then be paired with the exported figures for a complete training report.

12 Note on Console Log Availability

In the explicit run folder used above, matlab_console.log exists but is empty. The figure and policy output files are still valid and were used as primary evidence.